RealQM Chemistry: Introduction

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Abstract

RealQM offers a generalisation of Schrödinger's equation for the Hydrogen atom with $N = 1$ electron to atoms and molecules with $N > 1$ electrons, viewed as non-overlapping 3d unit charge densities subject to Coulomb interaction without self-repulsion and meeting at a common free boundary with continuity and zero normal derivative, thus in forms of classical multi-phase continuum mechanics in physical 3D space. RealQM offers a foundation of chemistry in terms of a parameter-free mathematical model expressed as an essentially 3-line computer code with linear computational complexity for ab initio simulation of molecular dynamics.

1 Mathematical Foundation of Chemistry

Chemistry as the science of *molecules* created by *chemical bonding* of *atoms*, seeks its foundation in *atomic physics* as textbook *Standard Quantum Mechanics StdQM* first expressed in mathematical form as a *Schrödinger Equation SE* in 1926 [2, 5, 7, 4, 10]. Major efforts over 100 years have been made to explain the physics of chemical bonding by StdQM, but there is still no consensus even for the simplest molecule as the *Hydrogen molecule* H_2 supposed to be formed by a *covalent bond* between two *Hydrogen atoms* H. Nordholm and Bacskay as leading theoretical chemists summarize the situation in 2020 [6]: "the concept of a covalent bond ... is still the subject of debate, disagreement, and ignorance with respect to its physical origin". The emergence of a discipline of *Philosophy of Chemistry* [3, 12] is an expression of unresolved foundational issues captured by Roald Hoffmann [1]: "What quantum mechanics gives us are numbers. But not stories.". StdQM thus appears as a black box delivering numbers but not physics-based explanations of what is inside the box.

In this note we present an alternative SE as the basis of *RealQM* as an alternative to StdQM, thus as an alternative computational foundation of chemistry offering "new numbers" together with "stories" as theoretical explanations. The basic concept of RealQM is *electrons* as *non-overlapping 3D unit charge densities* subject to *Coulomb interaction without self-repulsion* meeting at a common *free boundary* with *continuity and zero normal derivative*. This is a conception compatible with the clear ontology

of *classical multi-phase continuum mechanics* with RealQM as natural extension into atomic physics, which is computable in the same sense. RealQM supports a view of the dynamical evolution of a physical system as a form of *time-stepping analog computation* [8], which can be simulated in *digital computation* thus making *dynamical computational chemistry* foundational. RealQM can be implemented in a basically 3-line *explicit time-stepping* code capturing formation of molecules, which can be explained by theory. RealQM replaces the blackbox "Shut up and Calculate" of StdQM with "Calculate and Explain".

We start out returning to the critical moment in 1926 when Schrödinger having formulated his SE for H with one electron, faced the challenge to generalise SE to systems with $N > 1$ electrons. We shall see that the *formalist school* of Bohr-Born-Heisenberg-Dirac overpowered the *realist school* of Schrödinger-Einstein, and so declared the foundation of chemistry to be StdQM. No alternative was ever formulated and so chemists have for 100 years been left by physicists to make sense of StdQM, while physicists have shifted interest to subatomic physics. RealQM could have been formulated by Schrödinger in 1926, or later or by somebody else, but such an initiative apparently did not surface under the total control of StdQM as expressed by Nobel Laureate Gell Mann at a Nobel Conference in 1976: "Niels Bohr brainwashed a whole generation of theorists into thinking that the job (interpreting quantum theory) was done 50 years ago."

Chemistry was by Dirac assigned the new foundation of StdQM to be explored by chemists, but the task showed to be much more difficult than envisioned. The abstract non-physical nature of StdQM showed distance to the essence of chemistry as building concrete physical molecules. RealQM presents an alternative foundation without this gap.

2 Outline

We formulate the basic assumptions of RealQM as a parameter-free mathematical model within classical continuum mechanics, which we transform into a 3-line time-stepping code with work scaling with N. We give a survey of some of the capacity of RealQM in simulations/explanations of (1) why H_2 forms while He_2 and Au_2 do not, (2) how Helium He can combine with a proton H_+ to form the first molecule He_+H, (3) OrthoHelium vs ParaHelium, (4) the periodicity of the *Periodic Table PT* as an *electron packing problem*, (5) H_2O complemented by dynamical formation of a prototype $X3$ molecule , (6) extension to atomic nuclei and (7) existence/stability of matter. Complementing material is given on the RealQM web site [13] including articles [14] and code repository [16], together with many posts on the author's main blog [15].

3 Schrödinger 1926: H Atom with $N = 1$

Modern physics of QM was born in January 1926, when the 38 year old Austrian physicist Erwin Schrödinger after an outburst of creativity on vacation in the Swiss Alps together with a friend, rocketed to fame by formulating a mathematical model of the H

atom with one electron subject to *Coulomb attraction* from a *positive proton kernel* at $x = 0$ in the form of an *eigenvalue problem* for a *Hamiltonian* as a partial differential operator acting on a real-valued *wave function* $\Psi(x)$ depending on a spatial coordinate x in a 3D *Euclidean space* $\Re^3$, with $\Psi^2(x)$ representing *electron charge density*, named *Schrödinger's Equation SE*.

Schrödinger managed to compute the eigenvalues of SE analytically and found them to agree with the Balmer-Rydberg formula (1888) for the observed spectrum of H and so could reveal a deep truth about physics expressed in terms of eigenfunctions $\Psi(x)$ determined as stationary points of a *total energy* $TE(\Psi)$ given by

$$TE(\psi) = K(\psi) + PK(\psi) \equiv \frac{1}{2}\int |\nabla\psi(x)|^2 dx - \int \frac{\psi^2(x)}{|x|}dx \tag{1}$$

with normalisation $\int \psi^2(x)dx = 1$, where K is named *kinetic energy* and PK is *Coulomb kernel potential energy*. Stationarity is attained by Ψ satisfying with $W(x) = -\frac{1}{|x|}$

$$\mathcal{H}\Psi \equiv -\frac{1}{2}\Delta\Psi + W\Psi = E\Psi, \tag{2}$$

with $\mathcal{H}$ *Hamiltonian* and E *eigenvalue*, which has the form of *classical continuum mechanics* built from a classical Coulomb potential and a stabilising *Laplacian* giving rise to kinectic energy in a form of *localisation energy*. The ground state of H with minimal total energy can be described as a charge density localising optimally around the kernel to decrease potential energy subject to a localisation cost in the form of kinetic energy. In the present context with $\Psi^2(x)$ representing charge density, the kinetic energy represents the new physics of QM, but the terminlogy is misleading, since no motion of electronic charge is present. In any case, SE for H can be solved analytically and so can be viewed to be completely understood from both mathematical and physical point of view. Schrödinger identified the minimal eigenvalue to be $TE = K + PK = \frac{1}{2} - 1 = -\frac{1}{2}$ with eigenfunction $\Psi(x) = \frac{1}{\sqrt{\pi}}\exp(-|x|)$ with charge density decaying with $|x|$ from peak at $x = 0$. The balance of PK and K is expressed by the *Virial Theorem* as $K = -\frac{1}{2}PK$.

The complete success of SE for the Hydrogen atom with one electron, asked for a quick generalisation to an atomic system with $N > 1$ electrons. Schrödinger hesitated, but was overpowered by the easy catch suggested by Max Born and Werner Heisenberg in the form of a purely formal extension without physics, simply by adding a new coordinate and Euclidean space $\Re^3$ for each new electron leading to a wave function $\Psi(x)$ depending on the coordinate $x \in \Re^{3N}$ as *configuration space*. It only remained to give the formal SE with wave function $\Psi(x)$ over configuration space a physical meaning for $N > 1$. This showed to require abandoning classical physics as realities for quantum physics as probabilities of possibilities, which was a high price, but became the *Copenhagen Interpretation* based on Born's idea of $\Psi^2(x)$ as a *probability density* as textbook StdQM. Schrödinger supported by Einstein protested, but were both marginalized.

StdQM is viewed as the basic theory of atom physics and so also of chemistry as the science of molecules made of atoms. But there is a back side to this golden coin which has become increasingly evident with computation taking over practice: StdQM

as account of all possibilities comes with *exponential computational complexity*. This has demanded draconic ad hoc dimensional reductions with *Density Functional Theory DFT* the ultimate reduction into one common electron density depending on $x \in \Re^3$ as the only option for larger N based on StdQM, however with unclear physics inherited from StdQM.

4 RealQM for Atoms with $N > 1$

> If we now have to deal with N particles, then $\Psi(x_1, ..., x_N)$ is a function of N variables x_1,...,x_N over N 3D spaces $R_1, ..., R_N$. Now first let R_1 be identified with the real space $\Re^3$ and integrate over $R_2, ..., R_N$. Second, identify R_2 with the real space and integrate over $R_1, R_3, ..., R_N$ and so on. The N individual results are to be added after they have been multiplied by certain constants which characterise the particles. I consider the result to be the electric charge density in real space. (Schrödinger in a letter to Lorentz June 6 1926):
>
> An important historical point is to be noted here. It might have been thought that the correct generalization for N particles is to use N functions of one variable instead of one function of N variables. Such a "wrong turn" did not happen historically, which is, after all, remarkable. (Lieb-Seiringer in Stability of Matter 2005)

To extend SE for H to Helium He with $N = 2$, it is natural from mathematical point of view to split 3D space into two halfspaces meeting at a plane through the kernel and let each electron occupy one half space and meet with continuity of charge density at the diving plane. This is what RealQM does and it shows to give a ground state energy in agreement with observation. We shall see that the split of electrons in He carries through the PT as a "two-valuedness" replacing that of spin.

The same idea for $N > 2$, which is essentially that of the above two quotes, builds on a subdivision of 3D space into non-overlapping domains Ω_i with boundary Γ_i for $i = 1, ..., N$, each occupied by a non-zero electron charge density, which meet with continuity on the common boundary Γ as the union of all $\Gamma_i \bigcap \Gamma_j$ with $i, j = 1, ..., N$. We let $H^1(\Omega)$ denote the set of real-valued functions which are square integrable together with first partial derivatives over a domain Ω. RealQM thus builds on a natural mathematical conception of electrons as charge densities occupying non-overlapping regions in space and meeting with continuity, which can also be given a clear physical meaning in terms of classical continuum physics.

RealQM for an atom takes the following form restricting here to *ground state* with *minimal energy*: Find the function $\psi(x) = \sum_{i=1}^N \psi_i(x)$ in $H^1(\Re^3)$ with $\psi_i \in H^1(\Omega_i)$ and $\int \psi_i^2(x)dx = 1$, which minimizes the *total energy*

$$TE(\psi) \equiv K(\psi) + PK(\psi) + PE(\psi)$$

as the sum of *kinetic energy*:

$$K(\psi) = \frac{1}{2}\int_{\Re^3} |\nabla\psi(x)|^2\,dx,$$

attractive kernel potential energy:

$$PK(\psi) = \int_{\Re^3} W(x)\psi^2(x)\,dx,$$

and *repulsive electronic potential energy*:

$$PE(\psi) = \sum_i \int_{\Omega_i} \sum_{j\neq i} V_j(x)\psi_i^2(x)\,dx,$$

where $W(x) = -\frac{Z}{|x|}$ is the potential generated by a pointlike positive kernel of charge Z and $V_j(x)$ is the potential (modulo factor 2) generated by electron charge density j defined by

$$V_j(x) = \int_{\Omega_j} \frac{\psi_j^2(y)}{2|x-y|}dy \quad \text{for} \quad x \in \Re^3.$$

We understand that the kinetic energy appears as a measure of charge density variation in space and does not express any motion of charge in space, but rather appears as an analog to elastic energy. In particular, the ground state charge density has no time dependence, which means that the ground state does not radiate energy and so can persist over time. On the other hand, a radiating dipole in resonance with external forcing can be created as oscillation betweeen ground and excited state. The model contains *no electronic self-repulsion* since $j \neq i$ in the summation to $PE(\psi)$. This is an essential difference compared to StdQM/DFT, where effects of overlapping wave functions including self-repulsion need cancellation. The SE of RealQM is a non-linear equation over 3D physical space with a free boundary determined by charge density continuity.

Variation of $TE(\psi)$ over $\psi = \sum_i \psi_i \in H^1(\Re^3)$ with $\psi_i \in H^1(\Omega_i)$, leads to the following characterization of a minimiser $\Psi = \sum_i \Psi_i \in H^1(\Re^3)$ with $\Psi_i \in H^1(\Omega_i)$ and $\int \Psi_i^2(x)dx = 1$, satisfying for $i = 1, ..., N$:

$$\mathcal{H}_i\Psi_i \equiv -(\frac{1}{2}\Delta + W + 2\sum_{j\neq i} V_j)\Psi_i = E_i\Psi_i \text{ in } \Omega_i, \tag{3}$$

$$\frac{\partial\Psi_i}{\partial n} = 0 \text{ on } \Gamma_i, \tag{4}$$

where E_i is the Lagrange multiplier for electron charge $\int \Psi_i^2\,dx = 1$, and free variation of ψ_i over Ω_i imposes the homogeneous Neumann boundary condition (4) with n normal to the boundary. The boundary Γ appears as a *Bernoulli free boundary* combining a homogeneous Neumann condition and a Dirichlet/continuity condition. RealQM can be seen as a form of DFT [7] based on one-electron non-overlapping charge densities forming a total charge density. RealQM also connects to Bader's AIM [10] but differs by being based on charge densities rather than basins of kernel attraction.

5 3-Line Code with Physics

The ground state energy $TE(\Psi)$ of RealQM is computed by a gradient method including the following 3 steps:

1. Each electron seeks to minimize its contribution to total energy subject to Coulomb interaction with the other electronsr and the kernel as expressed by (3) and (4).
2. The potentials V_j are computed as solutions the Poisson equation $-\Delta V_j = 2\pi\psi_j^2$ in $\Re^3$.
3. The free boundary is updated in terms of *level set functions* $\chi_i(x)$ serving as characteristics functions of each Ω_i as solutions to level set equations with coefficients β_i depending on jumps of wave functions across common boundaries moving the free boundary in the direction of lower charge density and so towards continuity.

A Code Repository [16] contains benchmark problems computed by realising 1-3 in the form of explicit time-stepping of the following equations on a fixed 1 million mesh points spatial grid as a 3-line code: For $i = 1, ..., N$,

$$\dot{\Psi}_i + \mathcal{H}_i\Psi_i = 0 \quad \text{in } \Omega_i,\ \frac{\partial\Psi_i}{\partial n} = 0 \quad \text{on} \quad \Gamma_i \tag{5}$$

$$\dot{V}_i - \Delta V_i = 2\pi\Psi_i^2 \quad \text{in } \Re^3, \tag{6}$$

$$\dot{\chi}_i + \beta_i|\nabla\chi_i| = 0 \quad \text{in } \Re^3, \tag{7}$$

where the dot signifies derivation with respect to time, and in each time step Ψ_i is normalised to unit charge, as a dissipative process where molecular excess energy is absorbed by an environment. Computations typically are initialized with electrons well separated allowing the free boundary to develop dynamically with an overlap of level set functions of a couple of mesh points, and so the creation of a molecule can be followed. The codes in Code Repository can directly be tested and modified on the web.

RealQM directly extends to molecules by adding more kernels. Formation of a molecule can be computed by evaluating forces on kernels from electron charge potentials and moving kernels until equilibrium. More generally RealQM opens to *Molecular Dynamics MD* based on physical forces on kernels from electron potentials, instead of the energy based pseudo-forces using StdQM.

RealQM unifies computation and physics by viewing the 3-line code as a direct expression of dynamical formation of molecules under relaxation into balance of forces.

6 Summary RealQM

Electrons are viewed as non-overlapping unit charge densities interacting by Coulomb potentials and satisfying a Bernoulli free boundary condition. The above 3-line code executes time-stepping evolution towards minimal total energy as expression of the following physics:

- Each electron charge density is updated towards balance of potential and kinetic energies with potentials given by kernel and other electrons (5) under normalisation of charge densities including energy dissipation into environment.
- Electron potentials are updated (6).
- Free boundary is updated towards continuity of charge densities (7).

The code is parameter-free. The only user input is case specific data as kernel positions and number of electrons and grid resolution. Extension to MD is direct with forces computed as gradients of potentials.

Is it possible that RealQM can deliver anything new after a 100 year sleep to a field overloaded with complex quantum chemistry academic and commercial software (and scientifc reporting) asking for massive supercomputer resources now in competion with AI? Is it possible that RealQM can offer a new tool for ab initio simulation of *protein folding*, and if not, why? RealQM does not meet the difficulties of StdQM of *collapse of wave function*, *measurement*, *exchange*, *correlation* and *Pauli Exclusion Principle*, while not supporting *superposition*. This may seem as a death sentence, but is also a relief. Maybe after all, QM is not so strange? Maybe strange physics is not real physics. Maybe atom physics can be described by computable mathematics, like celestial mechanics described by Newton's Law of Gravitation. Maybe micro-physics has a simpler parameter-free strucure than macro-physics with many parameters building on micro-physics? Maybe chemistry can be seen as emergence of quantum physics revealed by computation. Maybe the need of a philosophy of chemistry can be reduced by clear ontology in computational mathematical form.

Computational forms of StdQM typically work with products of one-electron wave functions with overlapping supports as products of probabilities asking for compensation to capture *electron correlation*. RealQM computes with sums of one-electron wave functions as non-overlapping charge densities with full correlation.

Ground state energies of molecules are of prime concern in chemistry and are deterministically determined both in StdQM and RealQM. Comparisons can thus be focussed on computational complexity leaving StdQM interpretation without purpose.

7 Covalent Bonding

Chemistry can be seen as the physics of chemical bonding of atoms into molecules with lower energy than separate atoms, described as either *covalent* with "sharing of electrons" or *ionic* with "transfer of electrons". The unclear physicality of StdQM offers an unclear explanation of covalent bonding as (A1) localisation of electrons beween kernels decreasing potential energy and (B1) delocalisation of electrons over the whole molecule to avoid increase of kinetic energy from localisation [6]. But localisation and delocalisation appears to be contradictory from physical point of view and so (A1)+(B1) may not describe real physics and not even consensus.

RealQM explains covalent bonding as (A2) localisation of electrons beween kernels decreasing potential energy and (B2) free boundary allowing electrons to meet with non-zero charge and so without increase of kinetic energy from localisation. There is no contradiction between (A2) and (B2) and so (A2)+(B2) appears to represent physics better than (A1)+(B1). We first show RealQM applied to a 2d model molecule X_2 formed by a covalent bond between two Hydrogen like atoms X with one electron surrounding a kernel. We compute (code [25]) the total energy at different distances between the kernels to discover that the total energy decreases with decreasing distance until the kernel repulsion takes over with an increase of total energy. We analyse the contributions to total energy from kernel-electron, electron-electron and kernel-kernel potential energy and electronic kinetic energy to understand the dip to a minimum of total energy at a certain kernel distance forming the covalent bond.

We see in the screenshots of computations below in 2d wave functions (red) and 1d cross-cuts of wave functions (red), kernel potentials (blue) and electron potentials (light-blue). We identify in particular the shaded area between the blue and light-blue curves which combine with the red to give a net decrease of potential energy with negative kernel-electron energy dominating positive electron-electron energy. We see that the wave functions meet with non-zero value allowing kinetic energy to stay constant. In the plots, potential energy refers to the combined kernel-electron and electron-electron energy. It is natural to name the chemical bond formed by the meeting of electron wave functions at a free boundary between the kernels as a *covalent bond*, but then with new meaning compared to StdQM. Note that the Neumann condition on the free boundary is imposed weakly and so is not visible in the plot and compare with Fig. 10 where it is present.

Figure 1

Figure 2

Figure 3

We now turn to the basic example of formation of a H_2 molecule which can be followed by running the RealQM 3D code [26] to discover a stationary state after explicit time-stepping as a H_2 molecule at a kernel distance of 1.4 atomic units. We can follow charge concentration forming between the kernels with wave functions meeting at a midway plane orthogonal to the line between kernels, in a process without increase of kinetic energy just like in the model case above. RealQM thus delivers a covalent bond forming H_2.

The energy is reported here as the same bracket as for He below, and for the same reason. The two ways of imposing the exclusion at $N = 2$ give -1.1336 with the electrons

sharing one density and -1.2750 with them on disjoint half-spaces, against the reference -1.174476: a shared density under-binds by 3.5%, disjoint domains over-bind by 8.6%, and the reference lies inside the span of the model's own two treatments. The bracket is wider than helium's, and no dissociation energy is quoted, because a dissociation energy would assert a position inside it. (An earlier version of this passage quoted -1.17 Hartree, and a dissociation energy of 0.17 Hartree, as agreeing with the list values. That is withdrawn. The value came from a run on a much coarser mesh, $h = 0.13$–0.20, on which two large errors cancel: the pair repulsion is 7% low, which over-binds, while the coarse grid together with the explicit $r \to \sqrt{r^2 + h^2}$ kernel softening under-binds. Refinement removes the cancellation, and the axisymmetric reduction — which is exact for two domains on an axis and carries no angular discretisation error — converges the disjoint-domain construction to -1.276. So -1.17 is a value the relaxation passes *through* on its way to the construction's own minimum, exactly as -2.91 is for helium.) In the output plot the essence of the 3-line code is made visible.

With the same code (or directly [27]) we can follow the formation of Helium He letting the kernels coincide while neglecting kernel-kernel repulsion, which gives an energy of -2.86 with the two electrons sharing one density and -3.03 with them on disjoint half-spaces, bracketing the reference -2.9037 from either side. The two ways of imposing the exclusion at $N = 2$ fail in opposite directions — a shared density under-binds by 1.5%, disjoint domains over-bind by 4.2% — so the reference value lies inside the span of the model's own two treatments. (An earlier version of this passage quoted -2.91, within 0.24%, and said the agreement could be made exact by giving the kernel a small radius. Both are withdrawn: -2.91 is a value the relaxation passes *through* on its way to the construction's own minimum, and the kernel softening that reproduces -2.9037 scales as the mesh spacing, so it vanishes in the continuum and is a numerical constant rather than a physical kernel size.)

Having understood how two H atoms can form a H_2 molecule by a covalent bond as well as He, we now ask if He can similarly form a He_2 molecule. We run the code [28] to discover that this is not so, because the charge concentration between kernels is not taking place, because the two planes through the kernels separating the two electrons of each He atom, are pushed away from the center to maintain continuity of charge density at the separating planes. RealQM delivers the following total energies E for two He atoms at varying kernel distance D= 4, 3.2, 2.4, 1,6, E= 0.00, -0.04, 0.03, 0.20, showing very little bonding energy and so gives evidence that He does not form a He_2 molecule.

Figure 4: Non-existence of He2 molecule by non-accumulation of charge density between kernels by outward shift of planes P separating the electrons for each He atom to maintain continuity at P.

8 The First Molecule He_+H

Standard Cosmology says that Big Bang created free electrons and free protons forming neutrons together forming the Helium kernel from 2 protons and 2 neutrons, which then combined with electrons into the first atom as He, thus forming before combination into H, because of stronger kernel attraction. The first molecule was then formed by compressing a proton H_+ and He to form He_+H by transfer of an electron from He to H_+ in a weakly exothermic reaction, which can then dissociate into He_+ and H, thus altogether a process of forming H from He in the early universe [2, 9].

Figure 5: Helium atom with two half-space electrons and inserted proton

Figure 6: Formation of He_+H with one electron captured by proton.

We now let RealQM model this process (code [30]) to find He_+H. We see here the characteristic feature of RealQM for Helium with its two electrons occupying two separate non-overlapping half lobes meeting at a plane together with the +2 kernel in the middle, and we also see a proton inserted on the left. The energy of the system is the ground state energy E = -2.903 of He leaving the energy required for the proton insertion to the simulation. We watch the dynamics when the left electron is attracted to the proton and the right electron gradually takes over the Helium kernel ino a final state with energy E = -2.490 depicted in Fig. (6).Recall the shooting off an electron from He requires 0.903 Hartree to create He_+ which is a lot, while the above process with instead a proton capturing an electron involves only 0.413 Hartree from -2.903 to -2.490 with the rest borrowed from H.

9 Helium and OrthoHelium

A Helium atom in ground state has 2 electrons surrounding a +2 kernel. In StdQM the two electrons are overlapping with different spin. In RealQM electrons do not overlap (spin is not an issue) and so are restricted to two half spaces meeting at a plane through the kernel. The formation of the He ground state can be followed running the code [27] delivering a total energy of -3.03 Hartree with the electrons on disjoint half-spaces, against the reference -2.9037; the shared-density alternative gives -2.86, so the reference lies between the two. We can see He with its two halfspaces being formed by eliminating kernel repulsion in H2 letting the kernel distance go to zero.

Figure 7: Orthohelium as inner(red) and outer(green) shell electrons.

Figure 8: Orthohelium as excited half-space electrons.

He occurs in two excited forms as *OrthoHelium* with energy -2.175 and *ParaHelium* with slightly higher energy -2.148. We now let RealQM compute a state of He with one of the electrons excited to an outer shell with the remaining electron filling an an entire inner shell around the kernel. Running the code [17] we find that RealQM gives an energy of -2.175 ± 0.005 in agreement with Orthohelium, see Fig 7.

The observed spectrum of Orthohelium matches that of the theoretical spectrum of one electron atom with charge outside a sphere of radius about 0.3 au with kernel at the center of the sphere. This fits with the RealQM model of Orthohelium.

We compare using the code [18] excitation keeping the half-space geometry of the ground state to find again an energy of -2.175 ± 0.005 Hartree, see Fig. 8. RealQM thus delivers Orthohelium as the 1st excitation of He in two geometrically different settings.

StdQM presents Orthohelium as a complicated triplet state with the electrons having same spin, and Parahelium as a simple singlet state with the two electrons having opposite spin.

10 Periodic Table from Electron Packing

RealQM presents the ground state of an Atom as the result of an *electron packing* energy minimization problem, where each electron seeks to minimize its contributtion to the total energy under Coulomb attraction/repulsion from the atom nucleus and the other electrons, while the supports of the electrons meet at a common *free boundary* determined so as to maintain *continuity of electron charge density*. In this packing problem electrons take on a certain width/volume in 3D space determined by proximity to the kernel while competing for space with other electrons. RealQM can thus be seen as a "ballon packing" problem with electron kinetic energy serving as a measure of "compression". RealQM thus gives a rationale for the Aufbau of the PT, which is different from that of StdQM based on the s, p, d atomic orbitals of H [4].

RealQM packs electrons into an expanding sequence of on-overlapping spherical shells S_n centered at the kernel with $n = 1, 2, 3...$ with each S_n first subdivided into two half-spherical shells each of which is divided into two angular directions into $n \times n$ electron domains thus with a total of $2n^2$ electrons in each full shell S_n. This gives the familiar sequence 2, 8, 18, 32,... which is repeated in the periods of the Periodic Table 2, 8, 8, 18, 18, 32, 32.

In particular, RealQM suggests the following shell structures of filled shells with 8-shells radially divided into two 4-shells:

- (2)+(4+4)
- (2)+(4+4)+(4+4)
- (2)+(4+4)+(8)+(2)

- (2)+(4+4)+(18)+(2)
- (2)+(4+4)+(18)+(8)
- (2)+(4+4)+(18)+(18)+(4+4)
- (2)+(4+4)+(18)+(32)+(18)+(4+4)

Below we show computed ground state energies with code [23] (here reduced to 3 shells), where the electrons in each subshell are homogenised to spherical symmetry with the electron subshell structure indicated and table values in parenthesis. The shell configration is partly up to choice with best choice in principle having smallest energy, but the choice also has to be accessible in a dynamic build of the atom by succesively adding electrons. This opens a rich world to explore by RealQM.

- Lithium (2)+1: -7.55 (-7.48) 1st ionisation: (0.2)
- Beryllium (2)+(2): -15.14 (-14.57) 1st ionisation: 0.5 (0.35)
- Boron (2)+(2+1): -25.3 (-24.53) 1st ionisation: 0.2 (0.3)
- Carbon (2)+(2+2): -38.2 (-37.7) 1st ionisation 0.5 (0.4)
- Nitrogen (2)+(3+2): -55.3 (-54.4) 1st ionisation 0.5 (0.5)
- Oxygen (2)+(3+3): -75.5 (-74.8) 1st ionisation 0.5 (0.5)
- Fluorine (2)+(3+4): -99.9 (-99.5) 1st ionisation 0.5 (0.65)
- Neon (2)+(4+4): -132.4 (-128.5) 1st ionisation 0.6 (0.8)
- Sodium (2)+(4+4)+(1): -165 (-162)
- Magnesium (2)+(4+4)+(2): -202 (-200)
- Aluminium (2)+(4+4)+(2+1): -244 (-243)
- Silicon (2)+(4+4)+(2+2): -291 (-290)
- Phosphorus (2)+(4+4)+(3+2): -340 (-340)
- Sulphur (2)+(4+4)+(4+2): -397 (-399)
- Chlorine (2)+(4+4)+(3+4): -457 (-461)
- Argon: (2)+(4+4)+(4+4): -523 (-526)
- Calcium: (2)+(4+4)+(8)+(2): -670 (-680)
- Titanium: (2)+(4+4)+(10)+(2): -848 (-853)
- Chromium: (2)+(4+4)+(12)+(2): -1039 (-1050)

- Iron: (2)+(4+4)+(14)+(2): -1260 (-1272)
- Nickel: (2)+(4+4)+(16)+(2): -1516 (-1520)
- Zinc: (2)+(4+4)+(18)+(2): -1773 (-1795)
- Germanium: (2)+(4+4)+(18)+(2+2): -2089 (-2097)
- Selenium: (2)+(4+4)+(18)+(4+2):- 2416 (-2428)
- Krypton: (2)+(4+4)+(18)+(4+4): -2766 (-2788)
- Xenon: (2)+(4+4)+(18)+(18)+(4+4): -7355 (-7438)
- Radon: (2)+(4+4)+(18)+(32)+(18)+(4+4): -22800 (-23560)

Figure 9: Basic shell structure.

We see good agreement even with the relatively crude approximation of homogenisation into spherical symmetry used in the computations. In Figs. (10) we see interface and output for Oxygen in 2+3+3 configuration with 2+4+2 and 2+2+4 of similar energy.

To see the effect of the subshell structure we compare Neon: (2)+(4+4) with Neon: (2)+(8) without the (4+4) subshell structure, which has a ground state energy of -153, which is much smaller than the observed -128.5. We conclude that somehow the (4+4) subdivision is preferred before a subdivision without subshells. The difference between (8) and (4+4) is the homogeneous Neumann condition acting between subshells, tending to increase the width of the shell and thus increase the energy.

Figure 10: Atom Simulator Interface and Output.

The deeper reason for this preference remains to describe, but the intuition suggests that it relates to the shape or size of the domain occupied by an electron. With subshells electron domains are obtained by subdivision in both radial and azimuthal direction, while without subshells there is only azimuthal/angular subdivision of each shell.

We observe that ionisation energies, which are of similar size in different shells, become increasingly small as compared to ground state energies, and thus are delicate to compute as the difference between the ground state energies of atom and ion.

11 Approximate Shell Structure

RealQM shows an approximate electronic shell stucture for an atom with kernel charge Z to have *shell radius* $r_m \sim m^3/Z$ and *shell thickness* $d_m \sim m^2/Z$, where m is shell number with $2m^2$ number of electrons in filled shell. This conforms with charge density $\rho_m(r) \sim \frac{Z}{r^2}$, thus with a constant shell charge per unit of distance r to the kernel

in spherical symmetry, which gives a total energy $E \sim log(Z)Z^2$ in accordance with observations for $Z < 30$ outside first shell, and also for larger Z outside second shell. This also conforms to $d_m \sim \frac{1}{Z_{eff}}$ with Z_{eff} the effective charge for $r > r_m$.

The shell structure is determined from optimal *electron packing* under consecutive filling of shells, as a possible physical *design principle*.

12 From Weak to Strong Reacticity

A basic question of theoretical chemistry is why Gold Au has weak reactivity. To prepare an answer, we start recalling the following electron shell structure for atoms with one valence electron in the outermost shell including Au, with E 1st ionization energy according to NIST:

- E = 0.339 Au: 2+8+18+32+18+1
- E = 0.143 Cs: 2+8+18+18+8+1
- E = 0.205 Ag: 2+8+18+18+1
- E = 0.154 R : 2+8+18+8+1
- E = 0.213 Cu: 2+8+18+1
- E = 0.159 K: 2+8+8+1
- E = 0.188 Na: 2+8+1
- E = 0.198 Li: 2+1
- E = 0.500 H: 1

We see that Cs , Rb, K and Na all have shell structure ending with 8+1 together with observed small 1st ionization energy and strong reactivity, which can be explained by Coulomb attraction after transfer of the valence electron to form an ionic bond.

Figure 11: Two Au (H) atoms with large (small) inner shell radius and small (large) binding.

Figure 12

We see that Au, Ag and Cu all have electron structure ending with 18+1 together with observed large 1st ionization energy and weak reactivity decreasing with number of electrons. Large ionization energy prevents iconic bonds, but the possibility of a covalent bond remains. We now seek a theoretical explanation of the observed fact that Au does

not form an Au_2 molecule, while H forms H_2. What is the essential difference between Au and H both having one valence electron?

According to Standard Quantum Mechanics StdQM as the current canon, the weak reactivity of Au comes from a relativistic effect of inner electrons swirling around the kernel at half the speed of light thus gaining mass and so strongly binding the valence electron into weak reactivity. The same argument would seem to apply to Cs and even more to Fr and so contradict their strong reactivity. To explain the weak reactivity of Au as a relativistic effect does not seem to be real physics.

To seek to understand why Au does not form the molecule Au_2, while H forms H_2, we let RealQM using the code [24] compute the energy E1 of an X2 molecule formed by two X atoms with one valence electron outside a shell of radius R to get the following results with E1 minimal at a ceratin distance D1 compared with energy E2 at large distance D2:

- R = 0: E1 =-1.17 at D1 = 1.4, E2 = -1 at D2 = 4.
- R = 0.5: E1 = -0.87 at D1 = 2, E2 = -0.85 at D2 = 4.
- R = 1: E1 = -0.51 at D1 = 2, E2 = - 0.63 at D2 =4.

We make the following connection to H vs Au:

- R = 0: corresponds to H2 with energy release 0.17 under formation of molecule.
- R = 0.5: is intermediate with very small energy release 0.02.
- R = 1: corresponds to Au2 which needs input of 0.12 to form.

By inspecting the figures below it is easy to understand that (i) the covalent bonding of H_2 comes from presence of charge density between the kernels, (ii) the non-bonding of Au_2 comes from lack of space between the kernels for electron presence.

RealQM also offers an explanation of the large ionization energy of Au (0.339) compared to Cs (0.143) as an effect of the 18+1 shell configuration ending compared to 8+1. Ionization means that the valence electron is dragged away from the atom, which in the shell configuration of RealQM requires to also drag the 18 or 8 inner shell electrons to zero density with an increasing energy cost for 18. The even higher cost for H comes from the fact the the electron of H reaches the kernel, which the valence electrons of Au and Cs do not.

13 Water H_2O

Oxygen O has kernel charge +8 with the following possible electronic shell configurations 2+6, 2+4+2, 2+3+3 or 2+2+4. RealQM in spherical symmetry shows that both 2+4+2 and 2+3+3 match observation. We here choose 2+3+3 since it promotes a bent H_2O molecule. while 2+4+2 fits with CO2 with linear shape.

We next use the 3d code [19] to simulate the formation of a H_2O starting from a bent configuration matching 3 valence electrons in the outer shell of O to meet the valence electrons of H to form a covalent bond with dissociation energy matching observation, see Fig. 13.

Figure 13: H2O with O as 2+3+3

14 Molecular Dynamics

RealQM simulates the dynamical formation of a molecule moving each kernel according to forces computed as gradients of electron potentials and other kernel potentials until equilibrium with zero force. Running the code [29] we can follow the formation of a $X3$ molecule from 3 X atoms (2D prototype), each atom consisting of an inner shell with atom kernel shielded by electrons surrounded by an outer shell of valence electron charge density (red), which meet at a free boundary between the kernels. We see in Fig. (14) cross-cut through lower kernels of electron charge density (green), global potentials for lower inner shells (blue) with gradients (light-blue) indicating directions of update of kernel positions. We see kernels slowly moving in the directions given by light-blue gradients which tend to zero in approach to energy minimum. See also C_2H_4 [20] and C_2H_5OH [21].

Figure 14: Dynamic formation of $X3$ molecule.

15 RealQM for Nucleus

The molecule H_2 can be seen as two protons held together by Coulomb attraction from electron charge density accumulated between the protons, which overpowers proton-proton repulsion. Let us now see if a similar mechanism can keep together a nucleus consisting of an equal number of protons and neutrons, to give perspective on the balance of attractive and repulsive forces forming molecules.

The Nobel Prize in Physics 1963 was awarded to Maria Goeppert Mayer (1/4) and Hans Jensen (1/4) for a shell model of atomic nuclei with the protons and neutrons forming a nucleus arranged in shells in an attractive spherically symmetric potential. RealQM in the form of *RealNucleus* offers such a nucleus model with charge $+Z$ as a kernel of negative density of total charge $-Z$ surrounded by $2Z$ protons arranged in a shell system, viewing a neutron split into a proton and an electron. It is possible that RealNucleus connects to the magic numbers 2, 8, 20, 28, 50 associated with complete shells in the same way RealQM gives the periods of PT.

To see if such a model makes sense, let us check if Z negative point charges and $2Z$ positive point charges can form a stable configuration under Coulomb interaction. We

Figure 15: RealNucleus with 2 electrons surrounded by 4 protons

start with $Z = 1$ corresponding to 2H consisting of 2 protons surrounding a nucleus kernel of 1 electron. Suppose a linear particle configuration with the protons at coordinates -1 and +1 and the electron at 0. We have the following Coulomb potential energies:

- proton-electron attraction = -1-1 = -2
- proton-proton repulsion = +0.5
- total energy as (total potential energy)/2 = -0.75.

We find the total energy to be negative as evidence of stability/existence. We understand that this is a special case without electron-electron repulsion since self-repulsion is excluded.

We next consider $Z = 2$ corresponding to 4He consisting of 4 protons surrounding a nucleus kernel of 2 electrons, thus a case with non-zero electron-electron repulsion. Suppose a planar quadratic configuration with the electrons at (-0.5, 0) and (0.5, 0) and the protons at (-1.5,0), (1.5,0), (0,1.5) and (0.-1.5) in a 2d coordinate system, which gives the following Coulomb potential energies (with different spatial scale as compared to 2H)

- proton-electron attraction $< -1 - 1 - \frac{1}{2} - \frac{1}{2} - \frac{4}{1.5\sqrt{2}}$
- proton-proton repulsion $= \frac{1}{3} + \frac{1}{3} + \frac{4}{1.5\sqrt{2}}$
- electron-electron repulsion = 1
- total energy as (total potential energy)/2 $< -\frac{2}{3}$.

We find a total energy which is clearly negative even in the presence of electron-electron repulsion from the kernel. This is made possible by assuming a distance between the electrons in the kernel to be comparable to the distance to the protons, thus with a nucleus kernel of size comparable to that of the nucleus. We confirm stability using a RealNucleus code in spherical symmetry [22] with input/output in Fig. 15.

We thus find the enigma of the existence of a nucleus can be resolved as a marriage between the two Coulomb components of the system:

- control of electron-electron kernel repulsion by surrounding protons,
- control of proton-proton repulsion by electron kernel,
- a kernel of Z electrons binding 2Z protons,
- 2Z protons confining a kernel of Z electrons.

We find here a rationale for the near equal number of proton and neutrons in a stable nucleus. Altogether, RealQM + RealNucleus captures atom + nucleus in two steps:

- atom as electrons around a nucleus of protons of small size,
- nucleus as protons around a kernel of electrons of comparable size.

RealNucleus does not need the *strong force* of the *Standard Model*, which is remarkable.

16 Existence/Stability of Atoms and Matter

A lower bound on total energy of an atom guarantees *existence/stability*. The total energy of an atom with kernel charge Z is observed to scale with $-Z^\alpha$ with $\alpha \approx 1.81$, which is captured by RealQM with a suitable shell structure.

Existence/stability of *bulk matter*, as a collection of a large number M of atoms, is expressed by a lower bound on total energy scaling with $-M$ making it an *extensive quantity*. This is safely built in for RealQM since electrons do not overlap. The following Kato-Hardy inequality

$$\int \frac{\psi^2(x)}{|x|}dx \leq (\int \psi^2(x)dx)^{\frac{1}{2}}(\int |\nabla\psi(x)|^2dx)^{\frac{1}{2}}. \quad (8)$$

for all $\psi \in H^1(R^3)$, shows that potential energy is controlled in terms of kinetic energy, thus proving existence/stability.

A corresponding proof for StdQM is much more involved because electrons overlap, as evidenced in a 100 page proof by Dyson-Lenard later cut down by Lieb-Thirring. The fact that stability of matter is safely built into the structure of RealQM, can be seen as an expression of physicality of RealQM.

17 Spin

The *Pauli Exclusion Principle PEP* has no role in RealQM, which means that spin as basis for PEP has no role. But spin in StdQM has also a role in interaction with magnetic fields, and this role can be added to RealQM to capture e g the Stern-Gerlach experiment.

18 Molecule Model vs RealQM

Chemists use 3D Models of molecules formed by non-overlapping atomic charge densities with direct connection to spherically homogenized RealQM, see Fig. 16.

Figure 16: Balloon model vs RealQM BeH2 with Be 2+2 (hole + red/blue).

19 School Chemistry without StdQM

The lack of "stories" about chemical bonding based on StdQM means that school chemistry education is disconnected from StdQM, because stories/explanations are needed to give meaning for students. Chemical bonding is thus explained as a story of "sharing of electrons" in "Lewis structures" without connection to StdQM. RealQM offers stories/explanations based on physics understandable also by young minds, and so may have a role in school chemistry.

20 Conclusion

RealQM follows up the early idea of Schrödinger to view electrons as charge densities carrying identity from spatial occupation into a model of atomic physics within classical multi-phase continuum physics, which has linear computational complexity expressed in an essentially 3-line computer code. We show that RealQM captures basic aspects of atomic and molecular physics and so in particular may open new possibilities for ab initio protein folding.

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